$e_{123} = e_{231}$

a)

$$(\underline{a} \times \underline{b})_i = e_{ijk}\, a_j\, b_k = -e_{ikj}\, b_k\, a_j = [-(\underline{b} \times \underline{a})]_i$$

$$\underline{a} \times \underline{b} = -\underline{b} \times \underline{a}$$

$\alpha.\beta.\gamma = \gamma.\beta\alpha$

$b_j\, a_i\, e_{ijk}\, c_k = b_j\, a_i\, e_{jki}\, c_k = b_j e_{jki} c_k a_i$

b) $$\underline{a} \cdot \overbrace{(\underline{b} \times \underline{c})}^{\underline{d}} = a_i\, \overbrace{e_{ijk}\, b_j\, c_k}^{d_i} = b_j\, e_{jki}\, c_k \cdot a_i = b_j\, [\underline{c} \times \underline{a}]_j =$$

$$= \underline{b} \cdot (\underline{c} \times \underline{a}) \Rightarrow$$

$$\underline{a} \cdot (\underline{b} \times \underline{c}) = \underline{b} \cdot (\underline{c} \times \underline{a})$$

c) $$a \cdot (b \times a) = a_i\, e_{ijk}\, b_j\, a_k = b_j\, e_{ijk}\, a_i\, a_k = b_j\, e_{jki}\, a_k\, a_i$$

0

$$= b_j \cdot 0 = 0$$